

IoT (Internet of Things)

Time: 3 Hours

Max Marks: 75

Section – A

(Short Answer Questions – Answer any FIVE questions. Each carries 5 marks)

5 × 5 = 25 Marks

1. Define IoT. List and explain its characteristics.
2. Explain the architecture of IoT.
3. What is the role of sensors and actuators in IoT?
4. Describe the applications of IoT in healthcare.
5. What is MQTT? Explain its features.
6. Explain Cloud Computing and its relevance in IoT.
7. What are the security challenges in IoT?
8. Differentiate between IoT and M2M communication.

Section – B

(Essay Type Questions – Answer ALL questions. Each carries 10 marks)

5 × 10 = 50 Marks

9. (a) Explain the IoT layered architecture in detail.
OR
(b) Describe the various communication protocols used in IoT.
10. (a) Discuss different types of sensors used in IoT with examples.
OR
(b) Explain the working and application of RFID in IoT systems.
11. (a) What are the different network protocols used in IoT? Explain with examples.
OR
(b) Write about the role of Wireless Sensor Networks (WSN) in IoT.
12. (a) Discuss real-time applications of IoT in smart cities.
OR
(b) Explain the implementation of IoT in agriculture with suitable examples.
13. (a) Describe the security and privacy issues in IoT. How can they be addressed?
OR
(b) Explain the role of data analytics in IoT.